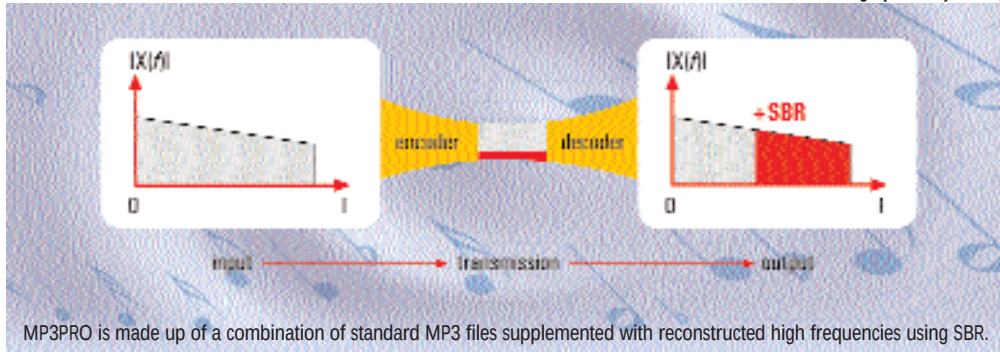




taken for the left and right audio channels. If we do the math, we get:

$44,100 \text{ samples per second} \times 16 \text{ bits per sample} \times 2 \text{ channels} = 1,411,200 \text{ bits per second}$ or $176,000 \text{ bytes per second}$

So a typical one-minute segment of CD audio occupies around 1MB of data. What MP3 does is reduce the number of bytes of a song while maintaining a high standard of audio quality. In order to accomplish this, the MP3 format uses a compression technique known as perceptual noise shaping or 'psychoacoustics'.

The principle behind perceptual noise shaping is based upon the characteristics of the human ear. For instance, the human ear is sensitive only to frequencies in the range of 20 Hz to 20,000 Hz. Anything below or above this range cannot be heard. Also, there are certain sounds that the human ear can hear much better than others. Furthermore, if two sounds of different amplitudes (volume levels) are being played simultaneously, the human ear has a natural tendency to pick up on the louder sound and focus

less upon the softer one. Using these principles, MP3 encoding will only include those frequencies, which the human ear 'prefers' and excludes certain parts of the song without significantly deteriorating sound quality.

With this algorithm of song compression, the resulting MP3 file is only one-tenth to one-fourth the size of the original song, giving it tremendous portability. What you have left after encoding is a 'near CD quality' song as the MP3 version of the song does not sound exactly like the original since some parts have been removed. For all practical purposes, the output is very close. Due to the fact that the MP3 encoding process eliminates some data from the original song, this type of compression is known as 'lossy compression'.

The MP3 format also supports Variable Bit Rate, or VBR for short. As mentioned earlier, MP3 files consist of hundreds of smaller chunks called frames. When encoding a VBR file, the encoder decides the bit rate for a particular frame. A frame's bit rate can drop down by 32

Lossy Compression

When audio is compressed, it usually doesn't mean the compression of the audio itself. What really happens is known as lossy compression. Lossy compression is usually used in audio, video and image files where certain details can't be seen or heard by the human senses and thus those areas are not included in the compressed file itself. Every human has a hearing range, which means that he or she can hear only a particular range of sounds. In MP3 compression or most audio format compressions, the frequencies that cannot be heard are eliminated from the particular song. However, a good audio compression method will give the most accurate results by getting most frequencies encoded within the song, giving quality of sound and also giving small file sizes.

A Question of St@NdaRDS

Sony, Liquid, Dolby. Three companies with three standards—AC3, ATRAC3, and LQT. Which is the best?

AC3: Audio Code number 3 is a sound compression standard that has been created by Dolby. In most audio formats, stereo sound is the standard. However, in AC3, the aim is to deliver five speaker streams and also a bass channel, which is overall denoted as 5.1 surround sound. This standard is usually found in cinemas and theatres. AC3 encoding allows sound from the frequency of 3 Hz to 20 kHz to play on the system, which is quite over the sounds that a human can hear. The bass that is produced is between the frequency ranges of 3 Hz to 120 Hz only. The bit rate at which the audio is played at is 640 kbps, which is much over most audio formats today like MP3, WMA, etc, but here quality and not the

media size is the main priority. Recently, with the introduction of DVD, compression and storage of surround sound is now possible. This enables users to listen to Dolby Digit surround sound at home.

Sony ATRAC3: Sony was the first company to introduce portable music with the unforgettable Walkman. Then came the Memory Stick Walkman, which was like a pen and stored songs on the memory. Sony has been using its own formats; there were two main technologies used—ATRAC and Magic Gate Memory. Later, a software called OpenMG was introduced which would convert the normal MP3 to the Sony format, ATRAC3, before copying it to the memory. However, the public wasn't quite happy with the conversion of formats.

Liquid Audio's LQT: Liquid Audio SP3 is the latest of all the formats available, and it

has been created with security of the media as the chief priority. Liquid Audio SP3 is also known as Liquid Audio Secure Player Platform. The Web site and company offer firmware updates for any later version. Claims have been made that Liquid Audio with its technology will be able to protect the copyrights of the artists. There are huge archives and online stores available for the format where people can use carts and select songs from the Internet through the authorised sites. Currently, Liquid Audio has device partners such as Palm, Sony, and Toshiba, to name just a few. There is also a software player which is available for download from the Liquid Audio site itself. The free version and the Pro version of the player allow you to write CDs and import music at high quality. Liquid Audio is also said to use AC3 Dolby digital Codec for its audio.